

hw2-230a-q6

February 16, 2026

0.1 Problem 6: Applied Problem - Teenage Gambling

This problem is adapted from *Linear Models with R* by Julian J. Faraway.

The dataset `teengamb` (from R package `faraway`) concerns a study of teenage gambling in Britain.

0.1.1 Tasks:

- Fit a regression model with gambling expenditure as the response and sex, socioeconomic status (on a percentile scale from 0-100), income (in pounds per week) and verbal score (correctly defined words out of 12) as predictors. Present the output of the summary command.
- What percentage of variation in the response is explained by these predictors?
- Which observation has the largest (positive) residual? Give the case number.
- Compute the mean and median of the residuals.
- Compute the correlation of the residuals with the fitted values.
- Compute the correlation of the residuals with the income.
- Fit a prediction interval for a new observation of a male teenager with a 60th percentile socioeconomic status, an income of 7 pounds per week, and a verbal score of 3 words out of 12.
- With all other predictors kept constant, what would be the difference in predicted gambling expenditure for a male compared to a female? Do you think it is reasonable to interpret this difference as a causal effect? Explain your reasoning.

```
[22]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
import statsmodels.api as sm
import rpy2.robj as ro
from rpy2.robj import pandas2ri
from rpy2.robj.packages import importr
from rpy2.robj.conversion import localconverter

faraway = importr('faraway')
teengamb_r = ro.r['teengamb']
```

```

# Convert to pandas dataframe
with localconverter(ro.default_converter + pandas2ri.converter):
    df = ro.conversion.rpy2py(teengamb_r)

print("Dataset Preview:")
print(df.head())
print("\n" + "="*80 + "\n")

print("Dataset Summary Statistics:")
print(df.describe())
print("\n" + "="*80 + "\n")

print("Dataset Info:")
print(df.info())
print("\n" + "="*80 + "\n")

print("Unique values in 'sex' variable:")
print(df['sex'].value_counts())

```

Dataset Preview:

	sex	status	income	verbal	gamble
1	1	51	2.0	8	0.0
2	1	28	2.5	8	0.0
3	1	37	2.0	6	0.0
4	1	28	7.0	4	7.3
5	1	65	2.0	8	19.6

=====

Dataset Summary Statistics:

	sex	status	income	verbal	gamble
count	47.000000	47.000000	47.000000	47.000000	47.000000
mean	0.404255	45.234043	4.641915	6.659574	19.301064
std	0.496053	17.262944	3.551371	1.856558	31.515866
min	0.000000	18.000000	0.600000	1.000000	0.000000
25%	0.000000	28.000000	2.000000	6.000000	1.100000
50%	0.000000	43.000000	3.250000	7.000000	6.000000
75%	1.000000	61.500000	6.210000	8.000000	19.400000
max	1.000000	75.000000	15.000000	10.000000	156.000000

=====

Dataset Info:

```

<class 'pandas.core.frame.DataFrame'>
Index: 47 entries, 1 to 47
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  -

```

```

0   sex      47 non-null    int32
1   status   47 non-null    int32
2   income   47 non-null    float64
3   verbal   47 non-null    int32
4   gamble   47 non-null    float64
dtypes: float64(2), int32(3)
memory usage: 1.7+ KB
None

```

```

=====

Unique values in 'sex' variable:
sex
0    28
1    19
Name: count, dtype: int64

```

```

[23]: # Part (a)
X = df[['sex', 'status', 'income', 'verbal']]
y = df['gamble']
X_with_const = sm.add_constant(X)

model = sm.OLS(y, X_with_const).fit()
print(model.summary())

```

OLS Regression Results

```

=====
Dep. Variable:          gamble    R-squared:                0.527
Model:                  OLS      Adj. R-squared:           0.482
Method:                 Least Squares    F-statistic:          11.69
Date:                   Mon, 16 Feb 2026    Prob (F-statistic):    1.81e-06
Time:                   14:37:49    Log-Likelihood:        -210.78
No. Observations:       47    AIC:                   431.6
Df Residuals:           42    BIC:                   440.8
Df Model:                4
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	22.5557	17.197	1.312	0.197	-12.149	57.260
sex	-22.1183	8.211	-2.694	0.010	-38.689	-5.548
status	0.0522	0.281	0.186	0.853	-0.515	0.620
income	4.9620	1.025	4.839	0.000	2.893	7.031
verbal	-2.9595	2.172	-1.362	0.180	-7.343	1.424

```

=====
Omnibus:                 31.143    Durbin-Watson:           2.214
Prob(Omnibus):            0.000    Jarque-Bera (JB):        101.046
Skew:                     1.604    Prob(JB):                 1.14e-22

```

Kurtosis: 9.427 Cond. No. 264.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Part (b): From the regression output in part (a):

- **R-squared = 0.527**
- **Adjusted R-squared = 0.482**

The four predictors (sex, socioeconomic status, income, and verbal score) explain 52.7% of the variation in teenage gambling expenditure. The remaining 47.3% of the variation is unexplained by the model and could be due to other factors not included (such as peer influence, family gambling habits, personality traits, etc.) or random variation. The adjusted R-squared of 48.2% accounts for the number of predictors in the model and penalizes for model complexity.

```
[24]: # Part (c)
residuals = model.resid
max_residual_idx = residuals.idxmax()
max_residual_value = residuals.max()

print(f"Case number with largest positive residual: {max_residual_idx}")
print(f"Residual value: {max_residual_value:.4f}")
```

Case number with largest positive residual: 24
Residual value: 94.2522

```
[25]: # Part (d)
residual_mean = residuals.mean()
residual_median = residuals.median()

print(f"Mean of residuals: {residual_mean:.10f}")
print(f"Median of residuals: {residual_median:.4f}")
```

Mean of residuals: 0.0000000000
Median of residuals: -1.4514

```
[26]: # Part (e)
fitted_values = model.fittedvalues
corr_resid_fitted = np.corrcoef(residuals, fitted_values)[0, 1]

print(f"Correlation between residuals and fitted values: {corr_resid_fitted:.10f}")
```

Correlation between residuals and fitted values: 0.0000000000

```
[27]: # Part (f)
corr_resid_income = np.corrcoef(residuals, df['income'])[0, 1]
```

```
print(f"Correlation between residuals and income: {corr_resid_income:.10f}")
```

Correlation between residuals and income: 0.0000000000

```
[28]: # Part (g)
new_obs = pd.DataFrame({
    'const': [1],
    'sex': [0],
    'status': [60],
    'income': [7],
    'verbal': [3]
})

prediction = model.get_prediction(new_obs)
pred_summary = prediction.summary_frame(alpha=0.05)

print(f"Predicted gambling expenditure: {pred_summary['mean'][0]:.2f} pounds_
↪per week")
print(f"\n95% Prediction Interval: [{pred_summary['obs_ci_lower'][0]:.2f},_
↪{pred_summary['obs_ci_upper'][0]:.2f}]")
```

Predicted gambling expenditure: 51.55 pounds per week

95% Prediction Interval: [0.69, 102.41]

```
[29]: # Part (h)
sex_coefficient = model.params['sex']
sex_pvalue = model.pvalues['sex']

print(f"Sex coefficient: {sex_coefficient:.3f}")
print(f"p-value: {sex_pvalue:.3f}")
```

Sex coefficient: -22.118

p-value: 0.010

With all other predictors held constant, males spend 22.12 pounds per week more than females on gambling. This difference is statistically significant at the common level of 0.05. I don't believe that it is reasonable to interpret this as a causal effect because this is observational data with potential confounding variables that differ between males and females and also affect gambling behavior. To establish causation, we would need either a randomized experiment (impossible for sex) or much stronger quasi-experimental designs that carefully account for all potential confounders.